# Linear Algebra – Problem Set 6 — Solutions

Eigenvalues & Eigenvectors · Full solutions · 6 short problems · Original problems in the spirit of MIT 18.06 (Strang)

Every eigenvector below is checked by multiplying it out, and every spectrum is checked against det — the product of the eigenvalues. Those two habits catch almost every mistake in this topic.

## Problem 1

$A = \begin{bmatrix} 3 & 0 \\ 0 & -1 \end{bmatrix}$; test three vectors; why $\mathbf{v} \neq \mathbf{0}$.

**(a)**

$$A\begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 3 \\ 0 \end{bmatrix} = 3\begin{bmatrix} 1 \\ 0 \end{bmatrix}, \qquad A\begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \end{bmatrix} = (-1)\begin{bmatrix} 0 \\ 1 \end{bmatrix}, \qquad A\begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ -1 \end{bmatrix}.$$

So $(1,0)$ has eigenvalue 3 and $(0,1)$ has eigenvalue $-1$. $(1,1)$ is not an eigenvector: $(3,-1) = c(1,1)$ would need $c = 3$ and $c = -1$ simultaneously. Eigenvectors: $(1,0)$ with $\lambda = 3$, and $(0,1)$ with $\lambda = -1$. **(b)** Because $A\mathbf{0} = \mathbf{0} = \lambda\mathbf{0}$ holds for *every* $\lambda$. If $\mathbf{0}$ counted, every number would be an eigenvalue of every matrix: there would be no finite list of $n$ eigenvalues, and $\det A$ = product of the eigenvalues would be meaningless. Requiring $\mathbf{v} \neq \mathbf{0}$ is what makes the spectrum a real piece of information.

## Problem 2

$B = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$: eigenvalues without a characteristic polynomial.

**(a)** Multiply the two symmetric directions:

$$B\begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 3 \end{bmatrix} = 3\begin{bmatrix} 1 \\ 1 \end{bmatrix}, \qquad B\begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix} = 1 \cdot \begin{bmatrix} 1 \\ -1 \end{bmatrix}.$$

$\lambda = 3$ with direction $(1,1)$, and $\lambda = 1$ with direction $(1,-1)$. **(b)** For a $2\times2$ matrix the characteristic polynomial has degree 2, so there are exactly two eigenvalues counting multiplicity — and the two we found are distinct, so there is no room for a third. The polynomial confirms it:

$$\det(B - \lambda I) = (2-\lambda)^2 - 1 = \lambda^2 - 4\lambda + 3 = (\lambda - 3)(\lambda - 1).$$

Product check: $\lambda_1\lambda_2 = 3 \cdot 1 = 3 = \det B$.

## Problem 3

$A = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$: the full computation.

**(a)** Only the diagonal changes when $\lambda I$ is subtracted:

$$A - \lambda I = \begin{bmatrix} 4-\lambda & 1 \\ 2 & 3-\lambda \end{bmatrix}, \qquad \det(A - \lambda I) = (4-\lambda)(3-\lambda) - 2 = \lambda^2 - 7\lambda + 10 = (\lambda - 2)(\lambda - 5).$$

$\lambda = 2$ and $\lambda = 5$. Check: $\lambda_1\lambda_2 = 2 \cdot 5 = 10 = \det A$ ✓. **(b)** Each eigenvalue gets its own reduction. $\lambda = 5$: $A - 5I = \begin{bmatrix} -1 & 1 \\ 2 & -2 \end{bmatrix} \to \begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix}$, so $x = y$ and $\mathbf{v} = t(1, 1)$. $\lambda = 2$: $A - 2I = \begin{bmatrix} 2 & 1 \\ 2 & 1 \end{bmatrix} \to \begin{bmatrix} 2 & 1 \\ 0 & 0 \end{bmatrix}$, so $2x + y = 0$, $y = -2x$ and $\mathbf{v} = t(1, -2)$. **(c)** Two multiplications:

$$A \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 5 \\ 5 \end{bmatrix} = 5 \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \qquad A \begin{bmatrix} 1 \\ -2 \end{bmatrix} = \begin{bmatrix} 2 \\ -4 \end{bmatrix} = 2 \begin{bmatrix} 1 \\ -2 \end{bmatrix}.$$

Both hold, so the pairing of eigenvalues and eigenvectors is correct.

## Problem 4

$M = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$: the zero eigenvalue.

**(a)** Row 2 is twice row 1, so $\det M = 0$ and $M$ is singular. A singular matrix has a non-zero vector with $M\mathbf{v} = \mathbf{0}$, i.e. an eigenvector with eigenvalue 0. For the other one, try the vector the first row suggests:

$$M \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 1 + 4 \\ 2 + 8 \end{bmatrix} = \begin{bmatrix} 5 \\ 10 \end{bmatrix} = 5 \begin{bmatrix} 1 \\ 2 \end{bmatrix},$$

so the second eigenvalue is 5. (Expanding $\det(M - \lambda I) = \lambda^2 - 5\lambda = \lambda(\lambda - 5)$ gives the same pair.) **(b)** For $\lambda = 0$, solve $M\mathbf{v} = \mathbf{0}$: $x + 2y = 0$ gives $\mathbf{v} = t(-2, 1)$. For $\lambda = 5$, $M - 5I = \begin{bmatrix} -4 & 2 \\ 2 & -1 \end{bmatrix}$ gives $y = 2x$, so $\mathbf{v} = t(1, 2)$. Check: $M(-2, 1) = (0, 0)$ ✓ and $M(1, 2) = (5, 10) = 5(1, 2)$ ✓. Eigenvalues 0 and 5; eigenvectors $t(-2, 1)$ and $t(1, 2)$.

## Problem 5

$S = \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix}$: repeated eigenvalue, one eigenvector.

**(a)** $S - \lambda I = \begin{bmatrix} 2 - \lambda & 1 \\ 0 & 2 - \lambda \end{bmatrix}$ is triangular, so $\det(S - \lambda I) = (2 - \lambda)^2$: $\lambda = 2$ with algebraic multiplicity 2. Now $S - 2I = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$, and $(S - 2I)\mathbf{v} = \mathbf{0}$ says $y = 0$ with $x$ free, so $\mathbf{v} = t(1, 0)$: the geometric multiplicity is 1. **(b)** No. Diagonalising a $2 \times 2$ matrix needs two independent eigenvectors to serve as the columns of $P$, and here the eigenspace is only the line $\text{Span}\{(1, 0)\}$. Equivalently: if $S = PDP^{-1}$ then both diagonal entries of $D$ would have to be 2, so $D = 2I$ and $S = 2I$ — false.

## Problem 6

$A = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$, $P = \begin{bmatrix} 1 & 1 \\ 1 & -2 \end{bmatrix}$, $D = \begin{bmatrix} 5 & 0 \\ 0 & 2 \end{bmatrix}$.

**(a)** Column $j$ of $AP$ is $A$ times column $j$ of $P$; column $j$ of $PD$ is $d_j$ times column $j$ of $P$:

$$AP = \begin{bmatrix} 5 & 2 \\ 5 & -4 \end{bmatrix}, \qquad PD = \begin{bmatrix} 5 \cdot 1 & 2 \cdot 1 \\ 5 \cdot 1 & 2(-2) \end{bmatrix} = \begin{bmatrix} 5 & 2 \\ 5 & -4 \end{bmatrix}.$$

They agree, so $AP = PD$. The equality is exactly the statement $A\mathbf{p}_1 = 5\mathbf{p}_1$ and $A\mathbf{p}_2 = 2\mathbf{p}_2$ — the columns of $P$ are eigenvectors — which is why no inverse is needed to recognise a diagonalisation.

**(b)** $(2,-1) = (1,1) + (1,-2)$, and $A$ multiplies the first piece by 5 and the second by 2 each time:

$$A^k \begin{bmatrix} 2 \\ -1 \end{bmatrix} = 5^k \begin{bmatrix} 1 \\ 1 \end{bmatrix} + 2^k \begin{bmatrix} 1 \\ -2 \end{bmatrix} = \begin{bmatrix} 5^k + 2^k \\ 5^k - 2^{k+1} \end{bmatrix}.$$

At $k = 1$: $(5 + 2,\ 5 - 4) = (7, 1)$, and indeed $A(2,-1) = (8 - 1,\ 4 - 3) = (7, 1)$ ✓.

Spotted an error, or a solution that skips a step? Open an issue on LA-hw and quote the problem number.